

CONTACT

Douglas Gordon Daly

Location: Portland, OR

Email: douglasgdaly@gmail.com

Phone: (650) 586-9720

LinkedIn: [linkedin.com/in/douglasdaly](https://www.linkedin.com/in/douglasdaly)

GitHub: github.com/dougaldaly

SUMMARY

Principal AI Engineer and Technical Leader with 20+ years of experience building systems that help organizations understand what is happening, why it is happening, and what to do next.

Expertise spans operational intelligence, decision-support systems, attribution analytics, observability, graph analytics, and AI-enabled business systems.

Designed and delivered solutions ranging from a patented radar resource-management system and enterprise-scale operational monitoring platforms to merchant intelligence systems, reliability analytics, graph-based dependency modeling, GenAI assistants, and AI-enabled decision support solutions. Combines deep expertise in machine learning, cloud platforms, data engineering, and modern LLM architectures with a proven ability to translate complex business problems into practical, measurable outcomes.

CORE TECHNOLOGIES

AI & Machine Learning: LLMs, RAG, Agentic AI, LangGraph, CrewAI, AutoGen, MCP, Prompt Engineering, PEFT/LoRA, Fine-Tuning, Diffusion Models, CLIP, Scikit-Learn, XGBoost

Cloud & Data Platforms: AWS, GCP, Vertex AI, BigQuery, SageMaker, Bedrock, S3, Neptune

Programming & Analytics: Python, SQL, Tableau, Pandas, NumPy, Jupyter, Elastic Stack, Splunk, Kafka

Data & Systems Engineering: Graph Analytics, Observability, Operational Intelligence, Entity Resolution, Data Modeling, Forecasting, Attribution Analytics

PROFESSIONAL EXPERIENCE

Principal Consultant | Daly Engineers LLC | Jan 2026 - Present

Consolidated Edison (via ABC Consulting)

- Built a governed analytics assistant using Gemini, BigQuery, and tool-calling architectures that applied analyst rules, operational controls, and auditable execution patterns for AI-enabled construction and financial analytics.
- Led analytical readiness assessment for Oracle/EBS and Maximo data migrated to BigQuery, establishing cross-domain analytics across construction operations and financial systems.
- Developed automated profiling, type inference, duplicate detection, candidate key analysis, and cross-system join validation frameworks across 100+ source tables and millions of operational and financial records.
- Designed peer-group and time-series anomaly detection readiness frameworks for construction actuals, including data suitability checks, gap analysis, nonzero filtering, and category-level cost behavior analysis.
- Developed a phased analytical maturity roadmap for anomaly detection, construction analytics, and AI-enabled decision support initiatives.

NBCUniversal (via Apex Systems)

- Built AWS-based AI and graph analytics pipelines using SageMaker, Terraform, S3, and containerized workloads.
- Developed graph-ready service dependency datasets from API telemetry and operational metrics to support digital twin initiatives.
- Evaluated graph analytics and graph machine learning approaches using Neptune, GraphStorm, Bedrock, and LangGraph for dependency analysis and operational intelligence.
- Authored architecture and analytical readiness assessments identifying data and operational gaps affecting long-term digital twin objectives.

AI Instructor | SimpliLearn & Interview Kickstart | March 2025 - Present

- Delivered AI and machine learning training across multi-month cohorts for software engineers, data scientists, and technical professionals through university- and enterprise-sponsored continuing education programs.
- Taught transformer architectures, LLMs, RAG, agentic AI, multimodal AI, MCP, diffusion models, enterprise AI governance, and practical AI development workflows.
- Developed hands-on demonstrations, Jupyter notebooks, and end-to-end applications covering modern AI workflows, evaluation techniques, and deployment considerations.
- Mentored students on AI architecture, governance, implementation tradeoffs, and enterprise adoption strategies.

Data Scientist | Intuit (via The Cydio Group) | Apr 2025 - Jul 2025

- Delivered analytics supporting business and operational decision-making for QuickBooks Capital.
- Investigated attribution anomalies across customer acquisition funnels, marketing campaigns, and conversion pathways to improve measurement accuracy and identify emerging business trends.
- Consolidated fragmented reporting into unified Tableau dashboards and automated executive reporting workflows using Python and SQL, reducing report preparation from days to hours.
- Partnered with finance, marketing, and product stakeholders to translate attribution and performance data into actionable business recommendations.

Data Scientist | Meta | October 2021 - January 2025

- Defined reliability KPIs and measurement frameworks for Meta FinTech, establishing leading indicators used to guide reliability planning, prioritization, and performance management.
- Developed forecasting and performance-management frameworks across business pillars and sub-pillars, translating engineering initiatives into projected reliability outcomes used for planning, goal setting, and leadership decision-making.
- Built attribution frameworks linking incidents, alerts, defects, and operational metrics to engineering organizations and service ownership structures, enabling more accurate accountability and performance measurement.
- Designed and developed the ReMon framework, enabling engineering teams to build comprehensive monitoring and alerting systems without requiring expertise in complex internal tooling; adopted across Meta FinTech, contributing to the automatic detection of more than 15 SEVs in 2024.
- Delivered impact sizing, ROI, and risk exposure analyses supporting investment and prioritization decisions across platform reliability, privacy compliance, authentication, testing infrastructure, and data platform initiatives.

Data Scientist | Samsung Electronics (via Harvey Nash) | Dec 2020 - Oct 2021

- Developed and deployed gradient-boosting propensity models used to target digital marketing campaigns, increasing campaign conversions by 146% and campaign-attributed revenue by 229%.
- Built customer sentiment analysis pipelines using NLTK and survey data to identify product and customer experience insights.

- Automated audience segmentation and campaign targeting workflows, improving efficiency and scalability across digital marketing initiatives.
- Improved marketing attribution and performance measurement capabilities through predictive modeling and analytics-driven decision support.

Head of Data Science | Facteus | Oct 2017 - Nov 2020

CONTEXT: Led data science and data-product initiatives on Facteus' 30B+ transaction dataset, the company's core source for market-intelligence products and revenue-generating analytics.

- Led data science initiatives on a 30B+ transaction dataset, creating analytical products and market-intelligence capabilities through large-scale feature engineering, merchant intelligence, and consumer-spending analytics.
- Developed merchant, brand, industry, and storefront resolution methodologies that transformed cryptic payment-card transactions into business-identifiable entities, enabling accurate market intelligence across more than 1,700 businesses with greater than 99% transaction coverage.
- Evaluated prospective data partnerships by assessing signal quality, coverage, and business value, supporting strategic investment and acquisition decisions.
- Produced transaction-based industry analyses during the COVID-19 pandemic, identifying shifts in consumer behavior, regional economic impacts, and e-commerce adoption trends before traditional reporting sources became available.
- Created data-driven reports and visualizations used by hedge funds, partners, and internal stakeholders to understand emerging market trends and business performance.

Expert SEO Solutions Architect | Nike (via Intersoft Inc.) | Nov 2016 - Sep 2017

- Designed a scalable SEO and information-architecture framework for Nike's product discovery experience, balancing customer navigation paths with search-engine indexation requirements across a rapidly expanding product catalog.
- Developed canonical URL, redirect, and XML sitemap strategies that improved search visibility while preventing duplicate content, redirect conflicts, and circular routing issues.
- Built Node.js proof-of-concept solutions demonstrating the feasibility and effectiveness of proposed SEO approaches, supporting adoption by business and engineering teams.

Senior Manager, Operations Analysis | Capital One | Jan 2012 - Dec 2016

- Developed predictive capacity-management models for customer-facing banking platforms, forecasting peak demand scenarios using customer-demand, temporal, and business-process drivers to support infrastructure planning and risk management.
- Identified critical gaps in enterprise monitoring operations after discovering that major incidents were frequently detected by customer call channels rather than operational monitoring systems.
- Conceived, designed, and led adoption of RITHM (Real-Time IT Health Monitoring), an operational intelligence platform that aggregated monitoring signals by business capability and system ownership, reducing major incident detection times from hours to minutes.
- Secured executive sponsorship for enterprise monitoring modernization, expanding initial Splunk-based prototypes into a scalable Kafka and Elastic Stack solution aligned with Capital One's open-source strategy.
- Partnered with CTO-level leadership to define enterprise monitoring strategy, operational health metrics, and reliability-improvement initiatives.
- Awarded the Capital One Business Innovation Award for contributions to operational intelligence and enterprise monitoring transformation; presented work at internal and external industry conferences.

Senior Principal Systems Engineer / Program Manager | Raytheon | Mar 2005 - Dec 2011

- Led systems engineering organizations of 40-50 engineers specializing in modeling, simulation, analysis, and advanced aerospace systems, providing technical leadership, career development, and proposal support for programs valued up to \$50M.
- Turned around an over-budget classified fixed-price technology program at approximately one-third completion by establishing execution plans, aligning engineering teams, and proactively managing technical and program risks; delivered the program under budget and nearly doubled the original profit target on a historically challenging delivery effort.
- Served as Principal Investigator for advanced signal-generation and radar resource-management algorithms, leading research, simulation, flight testing, and operational deployment across multiple tactical and surveillance aircraft programs.
- Developed a patented fuzzy-logic resource-management system (U.S. Patent, 2014) resulting from radar scheduling innovations deployed on major Department of Defense aircraft platforms.

SELECTED PROJECTS

GraphRAG Engine & LoRA Coverage Assistant | GitHub | 2025

- Built a GraphRAG system for multi-hop benefits reasoning with fully traceable retrieval paths across entities, clauses, and supporting evidence.
- Fine-tuned a LoRA adapter on a 7B open-source LLM to enforce a strict JSON schema (decision/reason/disclaimer) for reliable downstream automation.

Agent-to-Agent Protocol MVPs | GitHub | 2026

- Agent Olympics: built a small evaluation runner that benchmarks a generalist agent vs discovered specialists across three task categories with explicit scoring criteria and summary reporting.
- AI PEMDAS: built an instrumented coordination workflow where an evaluator routes sub-steps to ADD vs MULT agents and logs calls for end-to-end traceability (FastAPI JSON-RPC).

Agentic Multimodal Orchestrator | GitHub | 2025

- Orchestrated structured retrieval, layout planning, and diffusion prompting to generate data-driven posters/maps with repeatable, template-based workflows.
- Implemented template constraints and parameterized prompts to ensure consistency across runs.

EDUCATION

MBA, UCLA Anderson School of Management

Top Honors, GPA: 3.9

M.S., Applied Statistics, Rochester Institute of Technology

GPA: 3.9

M.S., Electrical Engineering, University of Washington

GPA: 3.7

B.S., Physics (Minor: Mathematics), Humboldt State University

GPA: 3.4

PATENTS & RECOGNITION

Lead Inventor, U.S. Patent #8635622

Method and System for Resource Management Using Fuzzy Logic Timeline Filling (2014)

Capital One Business Technology Innovation Award (2014)

Conference Chair & Speaker - Customer Analytics Innovation Summit (2016)

CAREER ARCHIVE

Career Stories

RAYTHEON

- Program turnaround
- Patent story
- Flight-test story

CAPITAL ONE

- July 7 capacity forecast
- Call center detecting SEVs
- RITHM creation and adoption

FACTEUS

- Merchant resolution challenge
- COVID economic impact analyses
- Data valuation / acquisition decisions

SAMSUNG

- Propensity model deployment
- 146% conversion increase
- 229% revenue increase

META

- ReMon
- Reliability forecasting
- Auto-detection KPI strategy

CONED

- Analytics assistant
- AI governance
- Explainable anomaly detection

NBCU

- Digital twin gap assessment
- Graph-based dependency modeling

Project Inventory

Raytheon

- Radar resource management
- Fuzzy logic scheduling
- Patent
- Flight test validation
- Fixed-price program turnaround

Capital One

- RITHM (Real-Time IT Health Monitoring)
- Enterprise monitoring modernization
- Capacity forecasting
- Splunk
- Kafka
- Elastic
- Business Innovation Award

Facteus

- Merchant resolution
- Store resolution
- Industry hierarchy mapping
- Data partnership evaluation
- COVID economic analyses

Samsung

- Propensity modeling
- Campaign targeting
- Sentiment analysis
- Attribution analytics

Intuit

- QuickBooks Capital
- Attribution analysis
- Funnel measurement
- Executive reporting automation

Meta

- Reliability KPIs
- SEV auto-detection
- ReMon
- Operational forecasting
- ROI assessments

NBCUniversal

- Digital Twin assessment
- Dependency graphs
- AWS
- Neptune
- GraphStorm

ConEd

- Oracle / Maximo integration
- Analytical readiness assessment
- Anomaly detection roadmap
- Guardrailed analytics assistant
- Gemini / Vertex AI / BigQuery

Technology Inventory

(Note: Should include technology, years & projects)

Early Career

- MATLAB
- C/C++
- Digital Signal Processing
- Modeling & Simulation

Analytics

- SAS
- SQL
- Tableau
- Python
- Statistics
- Forecasting

Data Platforms

- Kafka
- Elastic
- Splunk
- BigQuery

Cloud

- AWS
- GCP
- Vertex AI

- Bedrock

AI / GenAI

- OpenAI
- Gemini
- Hugging Face
- LangGraph
- CrewAI
- AutoGen
- MCP
- A2A
- RAG
- GraphRAG
- LoRA

Speaking & Teaching

Customer Analytics Innovation Summit (Chair)

Capital One internal conference presentations

External presentations on operational intelligence and monitoring

SimpliLearn AI/GenAI Instructor

Interview Kickstart AI Instructor

Awards & Recognition

Conference Chair & Speaker | Customer Analytics Innovation Summit (2016)

Capital One Business Technology Innovation Award (2014)

Patents

Lead Inventor, US Patent #8635622 | "Method and System for Resource Management Using Fuzzy Logic Timeline Filling" | Issued 2014

Certifications

Google Associate Data Practitioner

Google Ads Search Certification
(Expired)

Google Analytics GA4 Certification
(Expired)